Jeet Shah

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EDUCATION

The University of Sydney — Sydney, Australia
Master of Computer Science (Advanced Entry), Data Science & AI; Algorithms & Theory — *Aug 2026 - Present*

Birla Institute of Technology and Science, Pilani — Pilani, India
B.E. (Hons.) Computer Science; Minor in Data Science | CGPA: 8.0/10.0 — *Nov 2020 - Jul 2024*

EXPERIENCE

APTIV — Bengaluru, India
Advanced Software Engineer | C/C++, Python, Java/Android — *Jan 2024 - Jul 2026*

- Cut system failure-detection time by **40%** in system testing by building heartbeat-based health monitoring and diagnostics for distributed Android-Linux service workflows.
- Built production Android-Linux service interfaces using **gRPC/REST** and structured request/response flows, integrating vehicle functionality across heterogeneous software components.
- Extended **Android Audio / AudioControl HALs** for AAOS in C/C++ and Java (AIDL), implementing multizone routing, mmap audio paths and dynamic audio focus across media, navigation and alert streams.
- Built and evaluated real-time audio ML pipelines using **CNNs, LSTMs and spectrogram transformers**, reaching **85%** on IEMOCAP for emotion/context recognition.
- Developed a VAE-based adaptive soundscape achieving **20%** higher measured naturalness than static loops while maintaining **<100 ms** end-to-end latency.

Larsen & Toubro Technology Services (LTTS) — Mumbai, India
Data Science Intern | Computer Vision / OCR — *May 2023 - Jul 2023*

- Built a two-stage **P&ID detection + OCR** pipeline (overlapping crops, EAST, MMOCR) that improved baseline performance by **20%**; semantic feature extraction cut human classification error by **30%**.

RESEARCH EXPERIENCE

Pragya AI Lab, BITS Pilani — Remote
Research Contributor | PragyaVLA - Language-Conditioned Humanoid Motion — *Apr 2026 - Present*

- Investigated benchmark coverage for language-conditioned humanoid motion through literature review, benchmark experimentation and gap analysis; built **Hugging Face Spaces** and ran hands-on **MuJoCo / NVIDIA Omniverse / Isaac Sim** experiments.

Birla Institute of Technology and Science, Pilani — Hyderabad, India
Undergraduate Research Assistant | Physiological Signal Anomaly Detection — *Sep 2022 - Dec 2023*

- Proposed the **Alternative Robust Mahalanobis Distance (ARMD)** metric for physiological-signal anomaly detection, improving accuracy by **25%** over the classical Mahalanobis approach.
- Designed an **LSTM autoencoder** over 144-timestep ECG windows to detect and characterize four categories of temporal anomalies (Scikit-learn, Keras, TensorFlow).

PROJECTS

OptiRoute - TSP Algorithm Analyser | *Python, Optimization, Algorithms* | GitHub — Apr 2023 - Aug 2023

- Benchmarked **7 TSP strategies** (genetic algorithm, simulated annealing, PSO, dynamic programming, divide-and-conquer, greedy, brute force) with visual convergence analysis.

NanoGen - Mini C Compiler | *C/C++, Compiler Design* | GitHub — Jan 2023 - May 2023

- Built a mini C compiler covering **lexical, syntactic and semantic analysis**, symbol tables and intermediate-code generation for core C constructs.

Convex Hull + DCEL | *C++, Computational Geometry, Data Structures* | GitHub — Jan 2023 - May 2023

- Implemented **Graham-scan convex hull** construction and a **DCEL half-edge** data structure (vertices, half-edges, faces).

TECHNICAL SKILLS

Languages: C/C++, Python, Java, SQL, JavaScript, React
Systems / Backend: gRPC, REST APIs, Android, AAOS, Embedded Linux, AIDL, Docker, Git, Node.js
AI / Data: PyTorch, TensorFlow, scikit-learn, OpenCV, Pandas, NumPy
Certification: Bootlin - Embedded Linux Audio Development